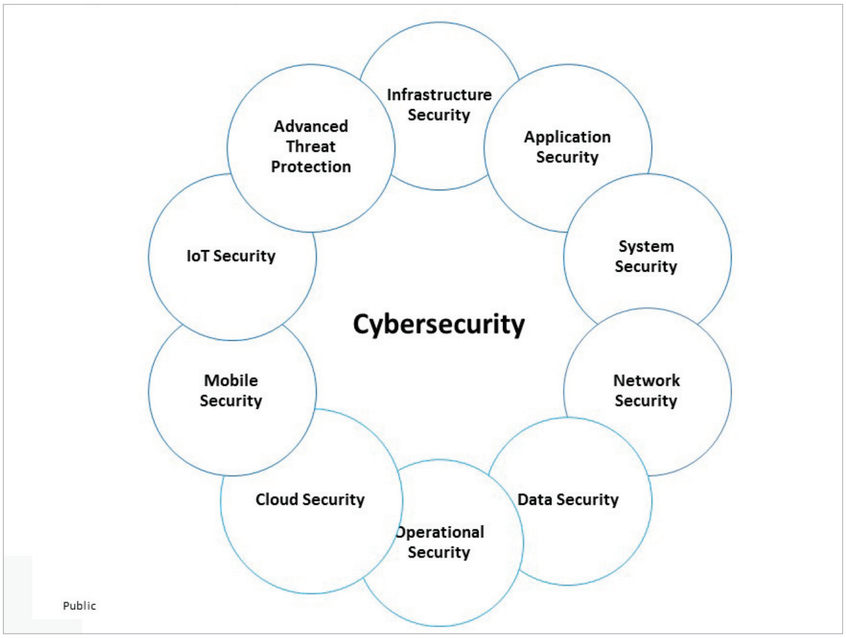

Figure 1: Types of cyber security

vulnerability and patch management

- Network security safeguards a computer network from intruders
- Data security, both in storage and in transport, which uses automated data encryption and data leakage prevention techniques
- Operational security refers to the procedures and actions used to manage and secure digital assets
- Cloud security needs to be continuously monitored and updated to safeguard data from attacks
- Mobile security helps in authentication and on-boarding, rogue access point detection, and managing wireless secure protocols
- Internet of Things (IoT) security protects products and services against threats
- Advanced threat protection covers botnet protection, malware analysis and anti-malware solutions, forensic solutions and automated security analysis

Industry trends in cyber security

According to a Grandview research report, the artificial intelligence component of the global cyber security market reached US$ 13.29 billion in 2021. It is expected to grow at a compound annual growth rate of 24.3 per cent from 2022 to 2030 to reach US$ 93.75 billion by 2030.

Mobile banking malware or assaults are on the rise (by 50 per cent), making portable devices a target for hackers. All our photographs, financial transactions, emails, and interactions put us in danger.

Another industry report says that organisations will be increasingly afraid to stack their security measures in 2023 because of these cyber security developments. They are likely to spend more than ever on asset protection this year, with estimates of US$ 100 billion or more.

Some key players operating in artificial intelligence in the cyber security market include:

- Acalvio Technologies, Inc.
- Amazon Web Services, Inc.
- Cylance Inc. (BlackBerry)
- Darktrace
- FireEye, Inc.
- Fortinet, Inc.
- IBM Corporation
- Intel Corporation
- LexisNexis
- Micron Technology, Inc.

Core concepts of AI and ML in cyber security

Artificial intelligence (AI) is a simulation of the human intelligence process by computers. It helps in identifying vulnerabilities, threats, and attacks in cyber space. This is the umbrella discipline under which fall machine learning and deep learning.

Machine learning (ML) uses existing behaviour patterns, making decisions based on past data and conclusions. ML consists of the programs developed to access the data stored on the system and make it more intelligent by studying the patterns and providing better support through continuous learning. There are three types of ML algorithms.

- ***Supervised learning:*** In this system both the input and the desired output data are provided. Input and output data are labelled for classification to provide a learning basis for future data processing. The term supervised learning comes from the idea that an algorithm is learning from a training data set, which can be thought of as the teacher. Data sets are labelled so that patterns can be detected and used to label new sets.
- ***Unsupervised learning:*** This involves the training of an AI algorithm using information that is neither classified nor labelled, and allowing the algorithm to act on that information without guidance. Data sets aren't labelled and are sorted according to similarities or differences.
- ***Reinforcement learning:*** This training method is based on rewarding desired behaviours and/or punishing undesired ones. Data sets aren't labelled but, after performing an action or several actions, the AI system is given feedback.